

08 — Visual inconsistency review & AI-generation assessment

What it detects: whether a page *looks* edited or synthetic to a vision model — inconsistent fonts, misaligned text, patchy colour/contrast, uneven sharpness or lighting — and whether it appears AI-generated. It is a **secondary, judgment-based** signal that can catch what pixel forensics miss, or corroborate them.

Code: `services/visual_inconsistency_service.py` (`run_visual_inconsistency_review`), `services/llm_service.py` (`analyze_page_visual_consistency` , `PageVisualAnalysis`), run by `tasks/visual_inconsistency_task.py` ; stores `check_type = visual_inconsistency_review` .

Algorithm

1. Render every page of the PDF (`render_pdf_pages` , 200 DPI) and encode as PNG, longest side capped at **1536 px** (`MAX_IMAGE_DIMENSION`). PNG, not JPEG, so glyph edges stay crisp and compression artifacts are not introduced.
2. Send each page to the vision model **twice, independently** (`RUNS_PER_PAGE = 2`), at `temperature = 0.4` — deliberately non-zero, unlike every other call in `LLMService` , so the runs are genuinely independent samples. Each run answers six structured questions: font consistency, text alignment, colour/contrast, resolution/sharpness, shadow/lighting — each a `VisualCategoryFinding` { `consistent` , `description` , `confidence` , `bounding_box` } — plus the `AIGenerationAssessment` { `likely_ai_generated` , `description` , `confidence` } .
3. **Corroboration rule** (the reliability mechanism): a category is reported as a real finding only if **both runs** flagged it, or **either** run flagged it with `high` confidence. A single low/medium one-off is kept but labelled "treated as noise" at severity `info` .
4. **Severity:** for a confirmed finding, `high` if both runs agreed *and* at least one was high-confidence, else `medium` ; everything not confirmed (both runs consistent, or an uncorroborated one-off) is `info` . The check's `result` is `flag` if any finding is `medium` or `high` , else `pass` .
5. The task appends `metadata_forensics_correlation` (info) cross-referencing the metadata result, so a reviewer sees editing-tool evidence next to visual evidence. It is **not scored** (avoids double counting), as is `experimental_signal_notice` (an info notice always added to the result).

Findings

finding	Meaning
<code>visual_font_consistency</code>	Fonts vary within the page
<code>visual_text_alignment</code>	Text or table alignment is inconsistent
<code>visual_color_contrast_consistency</code>	Colour/contrast differs across regions
<code>visual_resolution_sharpness_consistency</code>	Sharpness differs across regions
<code>visual_shadow_lighting_consistency</code>	Shadows/lighting inconsistent
<code>ai_generation_assessment</code>	The page may be AI-generated/synthetic
<code>experimental_signal_notice</code> , <code>metadata_forensics_correlation</code>	Info only, never scored

Output shape (real, truncated)

```
{
  "result": "flag",
  "details": [
    {
      "finding": "experimental_signal_notice",
      "severity": "info",
      "description": "This check uses a vision-capable language model's visual judgment, run twice independently..."
    },
    {
      "finding": "visual_font_consistency",
      "severity": "high",
      "page": 1,
      "description": "Page 1 – Font consistency: The font weight and style for the item descriptions ... This is a vision-model judgment, not pixel-level analysis – a probabilistic signal, not a definitive finding.",
      "bounding_box": {
        "page": 1,
        "x": 0.05,
        "y": 0.3,
        "width": 0.9,
        "height": 0.15
      },
      "data": {
        "category": "font_consistency",
        "run_agreement": "2-of-2 runs",
        "run_1": {
          "consistent": false,
          "confidence": "high",
          "description": "...",
          "bounding_box": {
            "x": 0.05,
            "y": 0.3,
            "width": 0.9,
            "height": 0.15
          }
        },
        "run_2": {
          "consistent": false,
          "confidence": "high",
          "description": "...",
          "bounding_box": {
            "x": 0.05,
            "y": 0.3,
            "width": 0.9,
            "height": 0.15
          }
        }
      }
    }
  ]
}
```

`data` keeps **both raw runs**, so a reviewer or developer can see exactly what was said and why a finding did or did not survive. The `bounding_box` (from the first run that supplied one) is drawn as a **dashed blue "AI-described area (approximate)"** overlay — deliberately styled to look less precise than the red/orange pixel-forensic boxes.

Known limitations

- **It is a language model's opinion.** Runs vary; corroboration reduces but does not remove hallucination. Boxes are approximate (models are poor at localisation).
- **AI-generation detection is unproven** on scanned/business documents. It is weighted lowest of any visual signal (weight 6) and every description says "experimental, probabilistic".
- **Every page costs two vision calls.** A 10-page PDF is 20 calls; there is no page cap on this task (unlike signature detection, capped at 10 pages).
- Legitimate documents made of mixed sources (a template with a pasted logo, mixed fonts) can flag.
- **Font judgement is unreliable:** the page is downscaled to 1536 px, and a few amounts retyped in a different font are often missed. The font consistency check covers fonts deterministically on digital PDFs, and with OCR font recognition on scans.
- PDF only.

Risk rules fed

Each rule fires only on `medium` / `high` findings (`severity_in`).

rule_id	Finding	Weight	Severity
<code>visual.font_inconsistency</code>	<code>visual_font_consistency</code>	12	medium
<code>visual.alignment_inconsistency</code>	<code>visual_text_alignment</code>	10	medium
<code>visual.color_contrast_inconsistency</code>	<code>visual_color_contrast_consistency</code>	12	medium
<code>visual.sharpness_inconsistency</code>	<code>visual_resolution_sharpness_consistency</code>	10	medium

rule_id	Finding	Weight	Severity
visual.lighting_inconsistency	visual_shadow_lighting_consistency	8	low
ai.generated_content_suspected	ai_generation_assessment	6	low